

class05_dataVisGgplot2

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Background

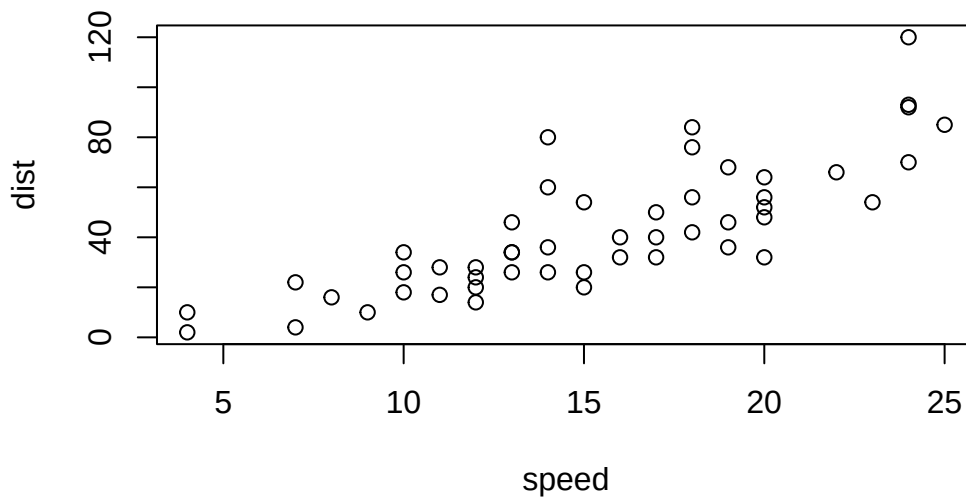
there are many graphics systems in R for making plots These include so-called “base” R graphics like the `plot()` function and add on packages like `ggplot2`.

We can use the in-built `cars` dataset:

```
head(cars)
```

```
  speed dist
1     4    2
2     4   10
3     7    4
4     7   22
5     8   16
6     9   10
```

```
plot(cars)
```

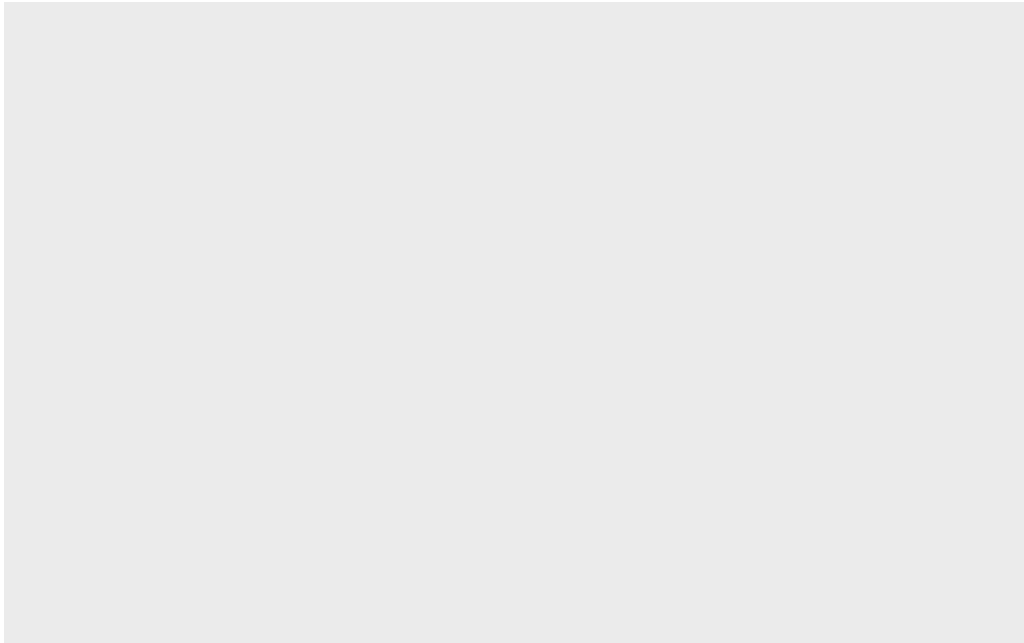


The main function in **ggplot2** packet is the `ggplot`, and before we run it we must install `ggplot2` by running `install.packages("ggplot2")` in the console

N.B. We never run `install.packages()` in our quarto doc because that would run it each time we render and become problematic

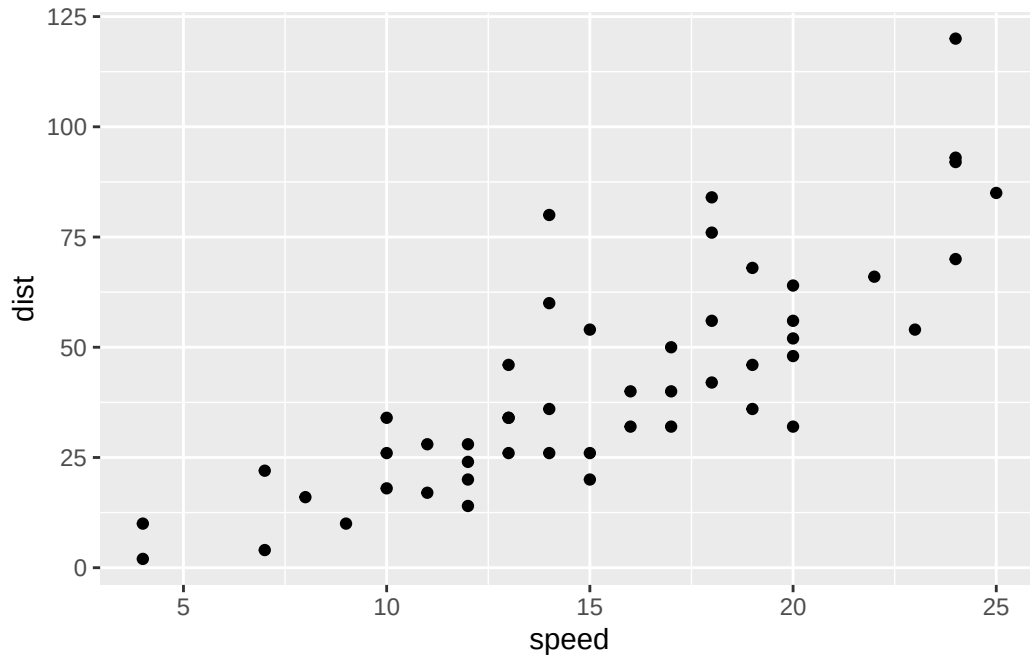
Once installed we need to load up the package into our R session

```
#install.packages("ggplot2")  
library(ggplot2)  
ggplot(cars)
```



Every ggplot has at least 3 layers: - The **data** (a data.frame of the stuff we want to plot) - The **aesthetics** (how the data maps to the plot) - The **geom** layer (how you want the plot drawn, e.g. points, lines, etc.)

```
ggplot(cars) +  
  aes(x=speed, y=dist) +  
  geom_point()
```

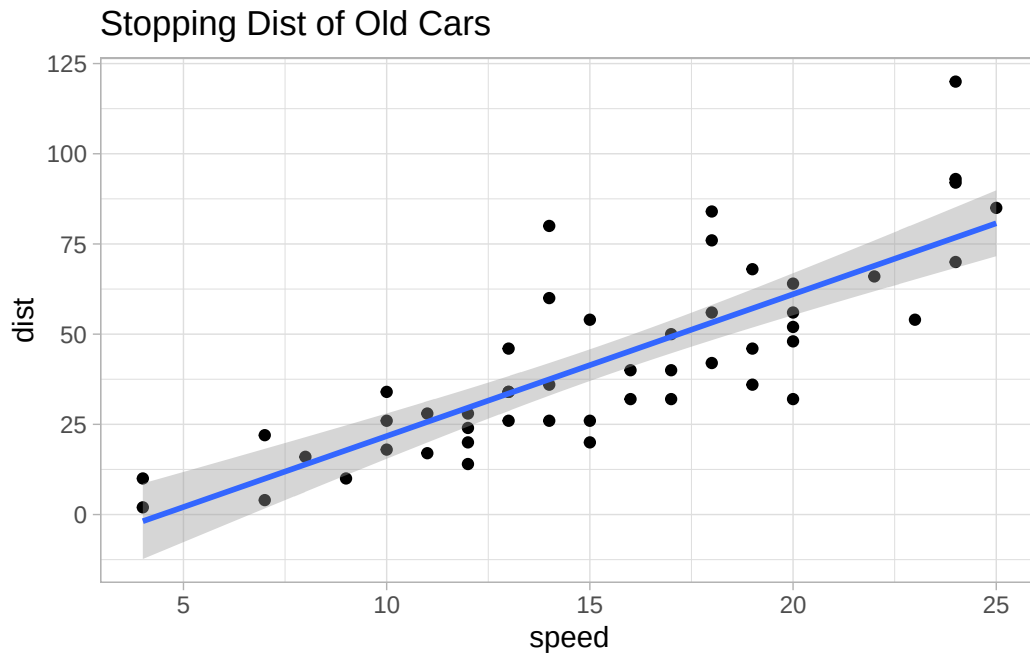


Add some custom features

Lets add a trend line that shows the relationship between speed and distance.

```
ggplot(cars) +  
  aes(x=speed, y=dist) +  
  geom_point() +  
  geom_smooth(method = "lm") +  
  theme_light() +  
  labs(title="Stopping Dist of Old Cars")
```

``geom_smooth()`` using formula = 'y ~ x'



Gene expression figure

```
url <- "https://bioboot.github.io/bimm143_S20/class-material/up_down_expression.txt"
genes <- read.delim(url)
head(genes)
```

	Gene	Condition1	Condition2	State
1	A4GNT	-3.6808610	-3.4401355	unchanging
2	AAAS	4.5479580	4.3864126	unchanging
3	AASDH	3.7190695	3.4787276	unchanging
4	AATF	5.0784720	5.0151916	unchanging
5	AATK	0.4711421	0.5598642	unchanging
6	AB015752.4	-3.6808610	-3.5921390	unchanging

```
sum(genes$State == "up")
```

```
[1] 127
```

```
genes[genes$State=="up",]
```

	Gene	Condition1	Condition2	State
10	ABCC3	0.93057376	3.260304	up
11	ABCC5	4.60042520	5.499443	up
420	AHNAK	6.62848950	7.611752	up
421	AHNAK2	3.65105100	5.181916	up
496	ALPPL2	0.55860496	2.625953	up
500	AMIG02	3.14356730	4.817252	up
503	AMOTL2	5.31663230	6.642013	up
525	ANXA2	7.52371000	8.284804	up
713	BCAS1	-3.68086100	1.604258	up
733	BMP7	-3.68086100	2.392754	up
736	BOC	5.79981700	6.891901	up
949	CACNG4	-0.06942624	2.863645	up
970	CAPS	3.18538760	4.542858	up
1050	CDH6	5.39595460	7.703885	up
1057	CDK6	5.73323700	6.704522	up
1062	CDON	4.42533870	5.571092	up
1067	CEACAM5	-3.68086100	3.571091	up
1068	CEACAM6	-3.68086100	4.114453	up
1134	CLU	2.79307000	4.210916	up
1151	COL1A1	4.87202170	5.865926	up
1158	COL9A3	0.93057376	3.114453	up
1168	COTL1	5.51280100	6.403785	up
1183	CRABP2	1.71906960	4.144826	up
1186	CRIM1	4.51553630	6.223422	up
1274	CTD-2033D15.1	0.05610465	2.845267	up
1333	CTGF	1.71906960	3.960744	up
1356	CYP1B1	4.62088900	5.582232	up
1358	CYP26B1	2.40187930	4.189221	up
1364	CYR61	4.65601730	6.492079	up
1461	DSCAM	-3.68086100	1.559864	up
1462	DSCAM-AS1	-3.68086100	4.872747	up
1492	EEF1A2	0.55860496	4.663152	up
1523	EMP1	6.03529170	7.417067	up
1536	EPHA5	3.91408560	5.221642	up
1546	ERBB4	3.14356730	4.367219	up
1659	FBLN1	4.80653240	6.489123	up
1661	FBN2	9.25945200	10.440404	up
1691	FHL2	2.17158170	3.636680	up
1705	FMN1	4.63101340	6.185574	up

1712	FOSL2	2.81099200	3.960744	up
1731	FXYD3	1.96299520	4.051717	up
1754	GATA3	-3.68086100	4.986129	up
1772	GFRA1	1.64106700	4.542858	up
1788	GLRA3	-3.68086100	1.466755	up
1819	GPC1	5.71906950	6.492079	up
1821	GPCPD1	4.64606760	5.854485	up
1822	GPFR	-1.94389530	1.986129	up
1834	GPR176	4.34150700	5.681397	up
1869	GXYLT2	3.02573100	4.537144	up
1875	H2AFJ	-3.68086100	1.986129	up
1922	HIST1H2BK	0.79307020	3.301331	up
1925	HIST2H4A	-1.52885800	3.083426	up
1926	HIST2H4B	-1.52885800	3.083426	up
1936	HMGA2	5.92235330	6.894138	up
1968	HSPG2	6.59138000	7.571790	up
1981	IER3	-1.20692980	3.002808	up
1996	IGFBP5	6.12756730	7.447085	up
1997	IGFBP7	4.84183000	5.984030	up
2086	KCNE4	-3.68086100	1.367219	up
2111	KIAA1324	-3.52885800	4.129720	up
2124	KIRREL	6.17850160	7.178250	up
2154	KRT5	-0.06942624	2.647327	up
2171	KYNU	-3.68086100	3.260304	up
2172	L1CAM	0.47114208	3.246365	up
2251	LIPG	2.35378530	4.239345	up
2253	LIX1	1.51553620	3.749689	up
2331	MALAT1	6.03147500	7.431769	up
2332	MALT1	7.36520200	8.369632	up
2391	MFSD2A	1.93057370	3.678805	up
2457	MSRB3	5.01417400	6.127821	up
2471	MTRNR2L8	4.28492300	6.191041	up
2472	MTRNR2L9	3.52642440	5.579455	up
2475	MUC1	3.10049870	4.253351	up
2480	MUC5B	-1.94389530	3.571091	up
2485	MYC	3.15764280	4.386413	up
2492	MYL9	5.56917430	6.705796	up
2537	NEAT1	1.79307010	4.218076	up
2605	NRP1	-2.52885800	1.769318	up
2618	NTS	1.86345960	5.817252	up
2635	OAS3	-3.68086100	1.417845	up
2684	OR52E8	-3.68086100	2.934904	up
2783	PDGFC	5.53723100	6.265497	up

2796	PEAK1	5.01803640	5.879537	up
2809	PGR	1.60042510	4.625953	up
2820	PIEZ01	4.69031050	5.998656	up
2859	PLK2	7.04384230	8.167194	up
2868	PMP22	3.83746430	4.994492	up
2924	PRICKLE1	3.04099750	4.328049	up
3039	RASGRP1	1.89740680	3.899714	up
3069	RERG	-3.68086100	2.083426	up
3109	RN7SL1	6.91822500	7.909146	up
3115	RND3	3.83746430	5.035598	up
3123	RNF144B	6.24921940	7.385618	up
3575	RP11-366L20.2	-0.20692979	2.788683	up
3990	RP11-738E22.2	4.70476200	5.863645	up
4034	RP11-79P5.2	0.64106710	3.159777	up
4292	RP5-977B1.11	4.60042520	5.749689	up
4325	RPS6KB1	5.59784650	6.399068	up
4339	S100A14	-0.94389530	3.678805	up
4340	S100A16	3.66096690	4.774183	up
4344	S100P	-3.68086100	1.917416	up
4389	SEMA3C	2.94687560	5.354280	up
4401	SERPINA3	-3.68086100	2.881792	up
4410	SETD7	4.67081450	5.531408	up
4430	SHISA3	0.71906954	2.969255	up
4483	SLC39A6	7.20838930	8.439064	up
4490	SLC6A14	-3.68086100	3.960744	up
4500	SLC7A2	3.07105500	6.067658	up
4544	SOX2	-3.68086100	2.260304	up
4557	SPDEF	-3.68086100	2.144827	up
4578	SPTSSB	-0.35893290	6.600154	up
4593	STARD10	3.37803270	4.417845	up
4624	SULF1	-2.52885800	2.392754	up
4625	SULF2	0.79307020	5.781451	up
4650	SYTL2	2.32912300	4.719736	up
4663	TAGLN	3.72853000	6.035598	up
4677	TBC1D9	4.68546100	5.639349	up
4719	TFF1	-3.68086100	4.840635	up
4732	THBS1	5.78629160	8.526009	up
4742	TIMP3	-1.52885800	3.647327	up
4801	TNNC1	1.60042510	3.788683	up
4880	TSPYL5	-3.68086100	2.314752	up
4960	VCAN	5.86774700	7.396703	up
4961	VCAN-AS1	1.68059550	3.328049	up
5027	XIST	1.71906960	3.514060	up

5092	ZIK1	-0.20692979	3.129720	up
5136	ZNF460	3.08585200	4.354280	up

A useful new function in this context is the `table()` function:

```
table(genes$State)
```

```

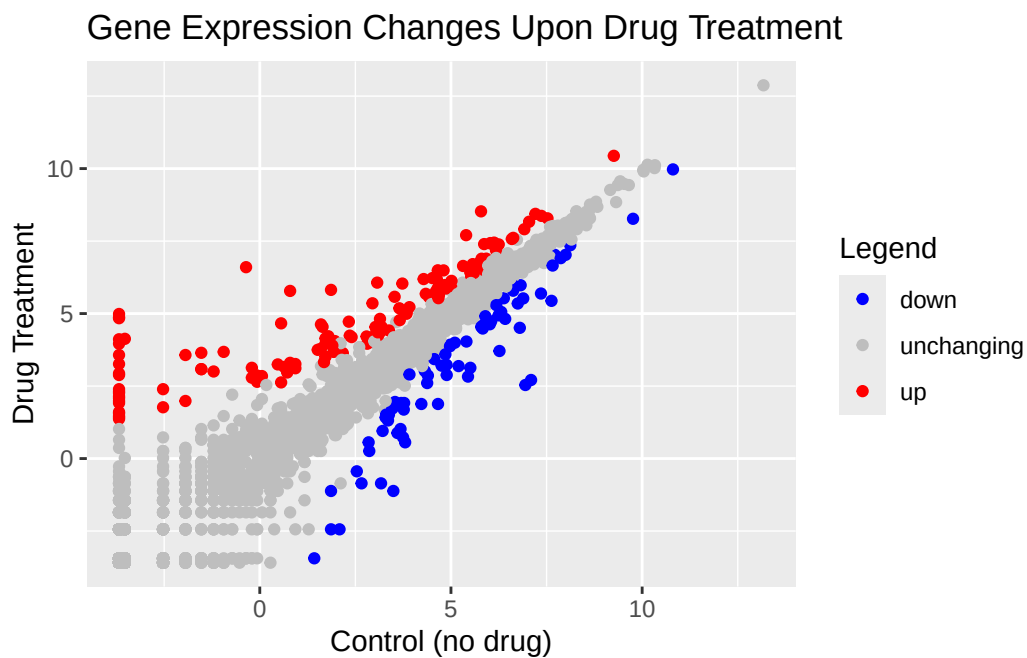
down  unchanging      up
  72      4997      127

```

```

ggplot(genes) +
  aes(x=Condition1, y=Condition2, color = State) +
  labs(title = "Gene Expression Changes Upon Drug Treatment", x = "Control (no drug)", y = "Drug Treatment") +
  scale_colour_manual( values=c("blue","gray","red") ) +
  geom_point()

```



Going further

Here we read the famous gapminder dataset

```
# File location online
url <- "https://raw.githubusercontent.com/jennybc/gapminder/master/inst/extdata/gapminder.tsv"

gapminder <- read.delim(url)
head(gapminder)
```

	country	continent	year	lifeExp	pop	gdpPercap
1	Afghanistan	Asia	1952	28.801	8425333	779.4453
2	Afghanistan	Asia	1957	30.332	9240934	820.8530
3	Afghanistan	Asia	1962	31.997	10267083	853.1007
4	Afghanistan	Asia	1967	34.020	11537966	836.1971
5	Afghanistan	Asia	1972	36.088	13079460	739.9811
6	Afghanistan	Asia	1977	38.438	14880372	786.1134

Q. How many entries (i.e. rows) are there in this dataset?

```
nrow(gapminder)
```

```
[1] 1704
```

Q. How many countries are in this dataset

```
length(table(gapminder$country))
```

```
[1] 142
```

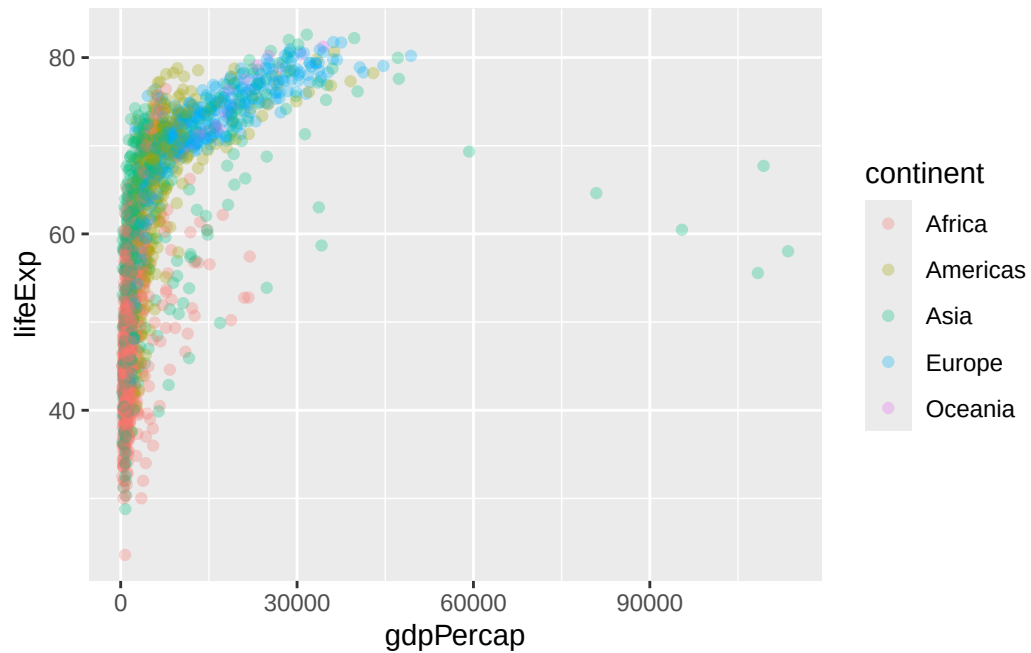
```
length(unique(gapminder$country))
```

```
[1] 142
```

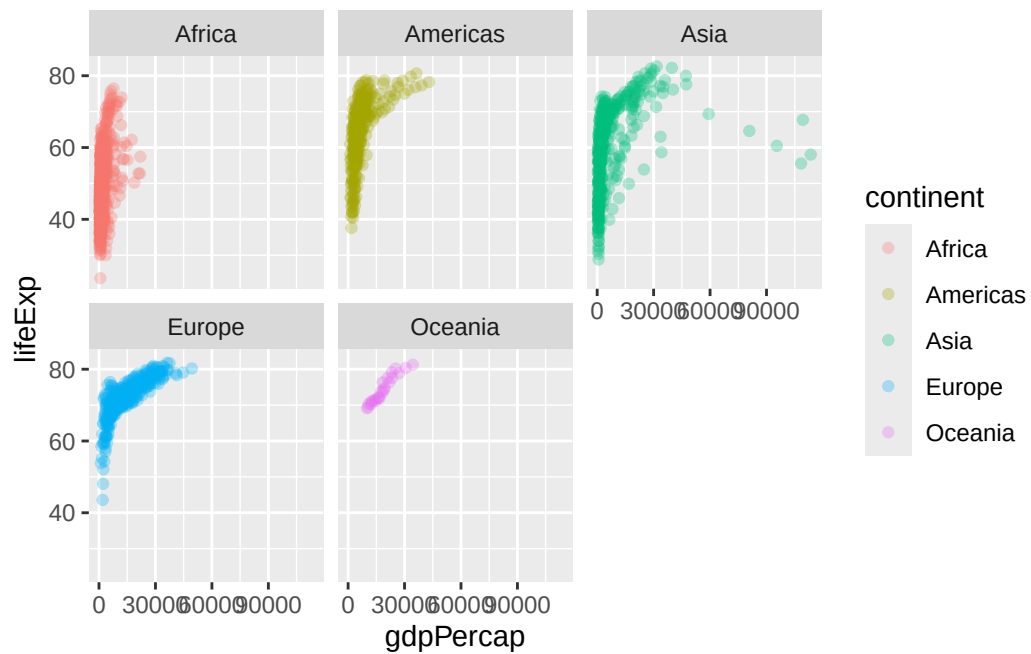
Let's make our first plot of the dataset:

```
p <- ggplot(gapminder) +
  aes(x = gdpPercap, y = lifeExp) +
  geom_point(aes(color=continent), alpha = 0.3)
```

```
p
```



```
p + facet_wrap(~continent)
```



Now we will make a plot for years 1977 and 2007

```
library(dplyr)
```

Attaching package: 'dplyr'

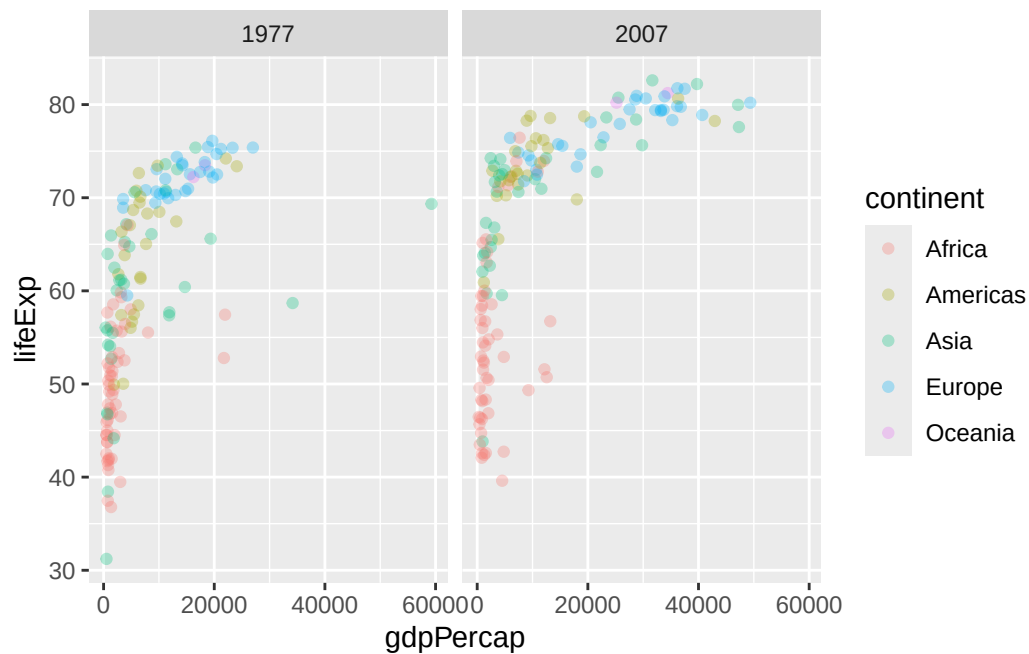
The following objects are masked from 'package:stats':

filter, lag

The following objects are masked from 'package:base':

intersect, setdiff, setequal, union

```
gapminder_1977_2007 <- gapminder %>% filter (year == 2007 | year == 1977)
ggplot(gapminder_1977_2007) +
  aes(x = gdpPerCap, y = lifeExp) +
  geom_point(aes(color=continent), alpha = 0.3) +
  facet_wrap(~year)
```

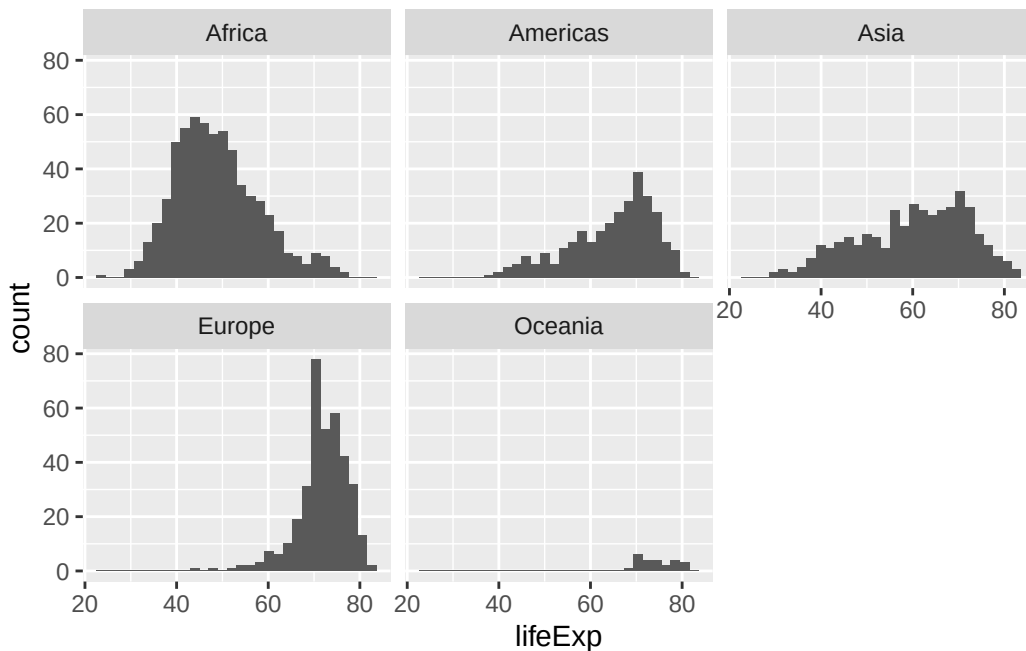


Q. Make a histogram of lifeExp faceted by continent and colored by continent

Faceted by continent:

```
ggplot(gapminder) +  
  aes(lifeExp) +  
  geom_histogram() +  
  facet_wrap(~continent)
```

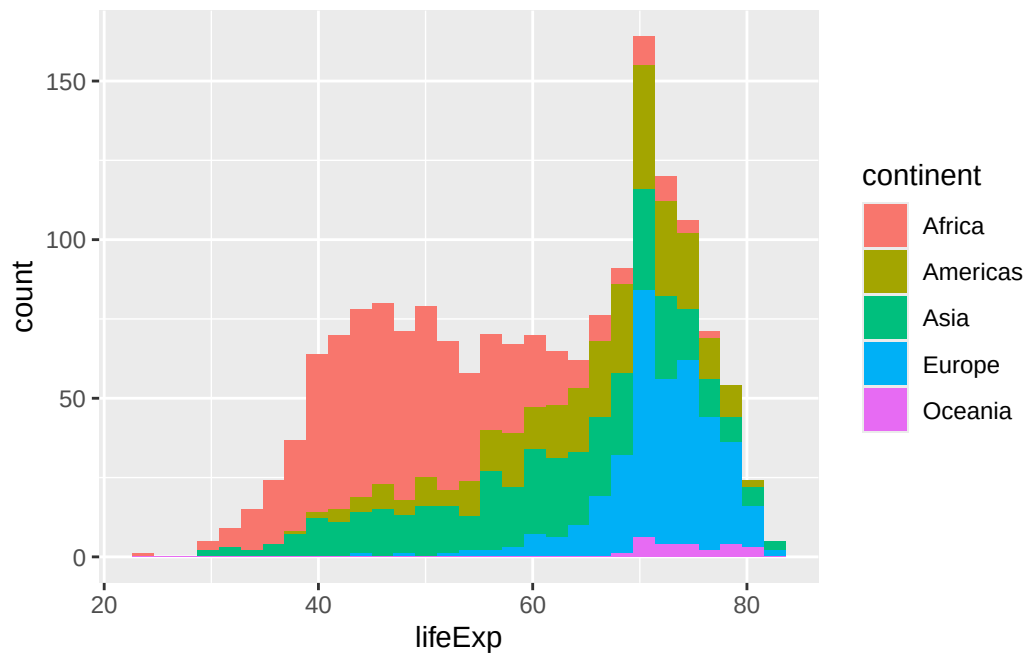
``stat_bin()`` using ``bins = 30``. Pick better value ``binwidth``.



Coloured by continent:

```
ggplot(gapminder) +  
  aes(lifeExp) +  
  geom_histogram(aes(fill=continent))
```

``stat_bin()`` using ``bins = 30``. Pick better value ``binwidth``.



Optional: Bar Charts!!

Organizing gapminder into the top 5 values of population in descending order in the year 2007

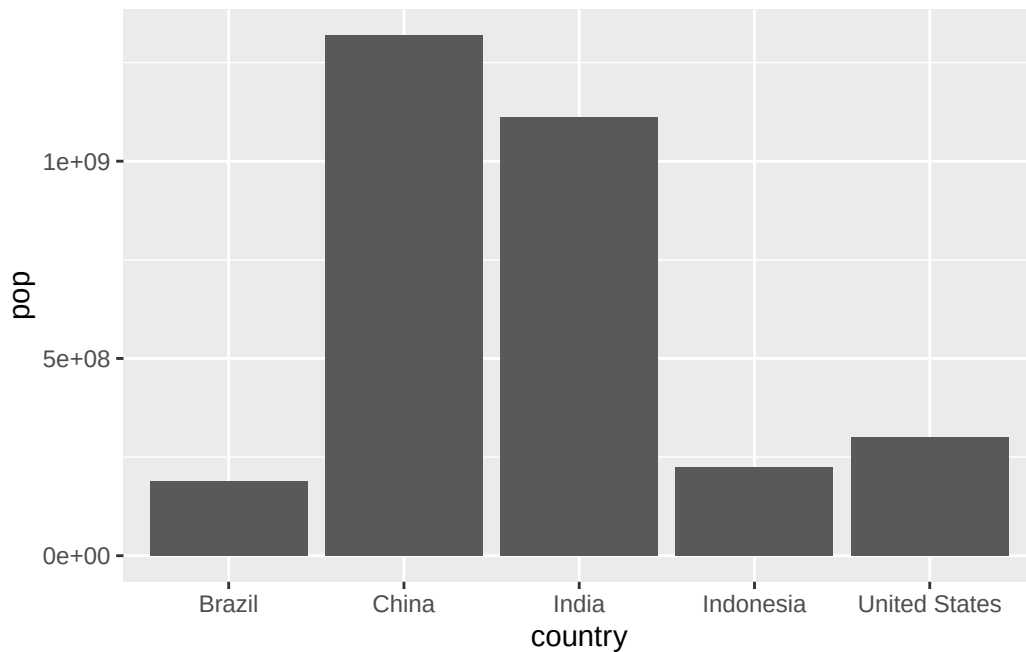
```
gapminder_top5 <- gapminder %>%
  filter(year==2007) %>%
  arrange(desc(pop)) %>%
  top_n(5, pop)

gapminder_top5
```

	country	continent	year	lifeExp	pop	gdpPercap
1	China	Asia	2007	72.961	1318683096	4959.115
2	India	Asia	2007	64.698	1110396331	2452.210
3	United States	Americas	2007	78.242	301139947	42951.653
4	Indonesia	Asia	2007	70.650	223547000	3540.652
5	Brazil	Americas	2007	72.390	190010647	9065.801

Creating the bar chart:

```
ggplot(gapminder_top5) +
  geom_col(aes(x = country, y = pop))
```



Q. How do you do this for life expectancy?

```
gapminder_top5_lifeExp<- gapminder %>%
  filter(year==2007) %>%
  arrange(desc(lifeExp)) %>%
  top_n(5, lifeExp)

gapminder_top5_lifeExp
```

	country	continent	year	lifeExp	pop	gdpPercap
1	Japan	Asia	2007	82.603	127467972	31656.07
2	Hong Kong, China	Asia	2007	82.208	6980412	39724.98
3	Iceland	Europe	2007	81.757	301931	36180.79
4	Switzerland	Europe	2007	81.701	7554661	37506.42
5	Australia	Oceania	2007	81.235	20434176	34435.37

```
ggplot(gapminder_top5_lifeExp) +
  geom_col(aes(x = country, y = lifeExp))
```

